Pass Task 2.1 - Travel Companion App

Task Description

Your task is to design and implement a Travel Companion App that helps international travellers convert essential values. The app must support conversions for Currency, Fuel Efficiency, and Liquid Volume.

Mandatory Requirement: Students must complete **all** subtasks assigned to their specific unit code to be eligible for a grade. Incomplete submissions will not be accepted.

Subtask 1: Design UI for the App (Mandatory)

The first subtask is to design the user interface (UI). The UI must include the following components:

- A dropdown menu or spinner to select the source unit/currency.
- A dropdown menu or spinner to select the destination unit/currency.
- A text field to enter the value to be converted.
- A button to initiate the conversion.
- A text view to display the converted value.
- *Optional*: Use icons or color-coding to distinguish between the three conversion categories.

Subtask 2: Implement the Conversion Logic (Mandatory)

Implement a function that takes the source unit, destination unit, and input value as parameters and returns the converted value. Use the following fixed conversion factors for this task:

Currency Conversions (Fixed 2026 Rates for Task)

- 1 USD = 1.55 AUD
- 1 USD = 0.92 EUR
- 1 USD = 148.50 JPY
- 1 USD = 0.78 GBP

Fuel Efficiency & Distance

- 1 Mile per Gallon (mpg) = 0.425 Kilometers per Liter (km/L)
- 1 Gallon (US) = 3.785 Liters
- 1 Nautical Mile = 1.852 Kilometers

Temperature (Climate Check)

- Celsius to Fahrenheit: $F = (C \times 1.8) + 32$
- Fahrenheit to Celsius: $C = (F - 32)/1.8$
- Celsius to Kelvin: $K = C + 273.15$

Subtask 3: Research on On-Device AI (Gemini Nano) (Mandatory)

In this subtask, you are required to research Gemini Nano (Google's efficient model for on-device tasks) and its use cases in Android development. As mobile hardware evolves, running AI locally is becoming standard. Write a 500-word report about Gemini Nano, including five specific ideas on how on-device AI can improve user privacy or offline functionality in mobile apps.

Additional sub-task (Mandatory) for SIT708 students

SIT708 students must complete all four subtasks (1, 2, 3, and 4) to meet the mandatory requirements. *(SIT305 students receive no extra marks for Subtask 4).*

Subtask 4: Add Validation and Error Handling

Add robust validation to ensure the app is "crash-proof":

- Prevent the app from processing empty or non-numeric inputs (show a Toast or Error message).
- Handle "Identity Conversions" (e.g., converting USD to USD) by notifying the user or returning the same value.
- Ensure the app handles negative values appropriately (e.g., fuel efficiency cannot be negative).

Submission Details: Task 2.1 - Travel Companion App

Note: You must follow the exact steps outlined below. Failure to comply with these formatting requirements may result in your task being marked as incomplete.
Responsible Use of AI: The use of Large Language Models (LLMs) is allowed only if declared.
Final Documentation Requirements
Your final submission must be a single **Word or PDF document** containing clear, **clickable links**. The document must include the following items: 1. LLM Declaration statement, 2. Research report, 3. Youtube link, 4. Github link, 5. LLM conversation link

1. LLM Declaration Statement

- A formal statement declaring whether or not you used a Chat LLM (e.g., ChatGPT, Gemini, etc.) to assist with this task.
- If AI was used, clearly list which parts of the code or research report were influenced by the AI.

2. Subtask 3: Gemini Nano Research Report (500 Words)

- **The full text of your 500-word report** must be included directly in this submission document.
- The report must cover the research into **Gemini Nano** and include five specific ideas for on-device AI use cases in mobile apps.

3. YouTube Demonstration Video Link

- **Duration:** 4–5 minutes (Strictly **not more than 6 minutes**).
- **Content:** Explain the logic of your conversion functions and your UI structure. You must walk through your code and explain the specific actions it performs.
- **Demonstration:** Show the app running on a device/emulator, demonstrating all conversion categories (Currency, Fuel, Temperature).
- **Quality:** The video must have a clear visual of the student (webcam), the code, and high-quality audio.

4. GitHub Repository Link

- Upload your entire Android Studio project to GitHub.
- The repository **must be set to Public**.
- The repository must include a **detailed README.md file**.

5. LLM Conversation Link (If Applicable)

- Provide a direct, clickable link to the exact chat history between you and the LLM.
- Include the **exact prompt** given and the **reasoning** for the use of that exact prompt for that specific component.

Final submission is due by **8:00 PM AEST, Friday, Week 4.**

File name format: Pass_Task2.1_YourName_StudentID.pdf while submitting to OnTrack